# HEMOGLOBINOPATHIES

# Learning Outcomes

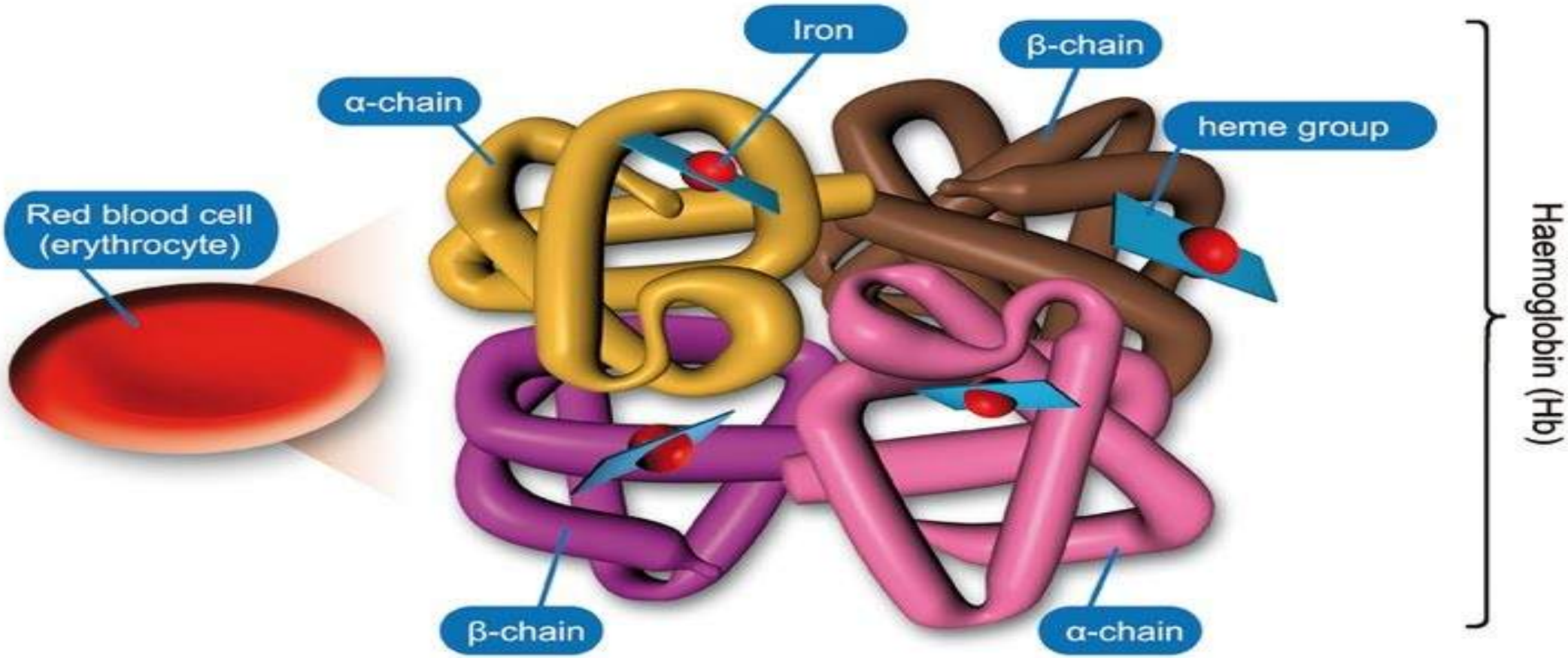
At the end of this session student will be able to understand;

1. Structure and functions of hemoglobin (normal mechanism).
2. Abnormalities linked with of hemoglobin.
3. Types / classification of hemoglobinopathies.

# HEMOGLOBIN

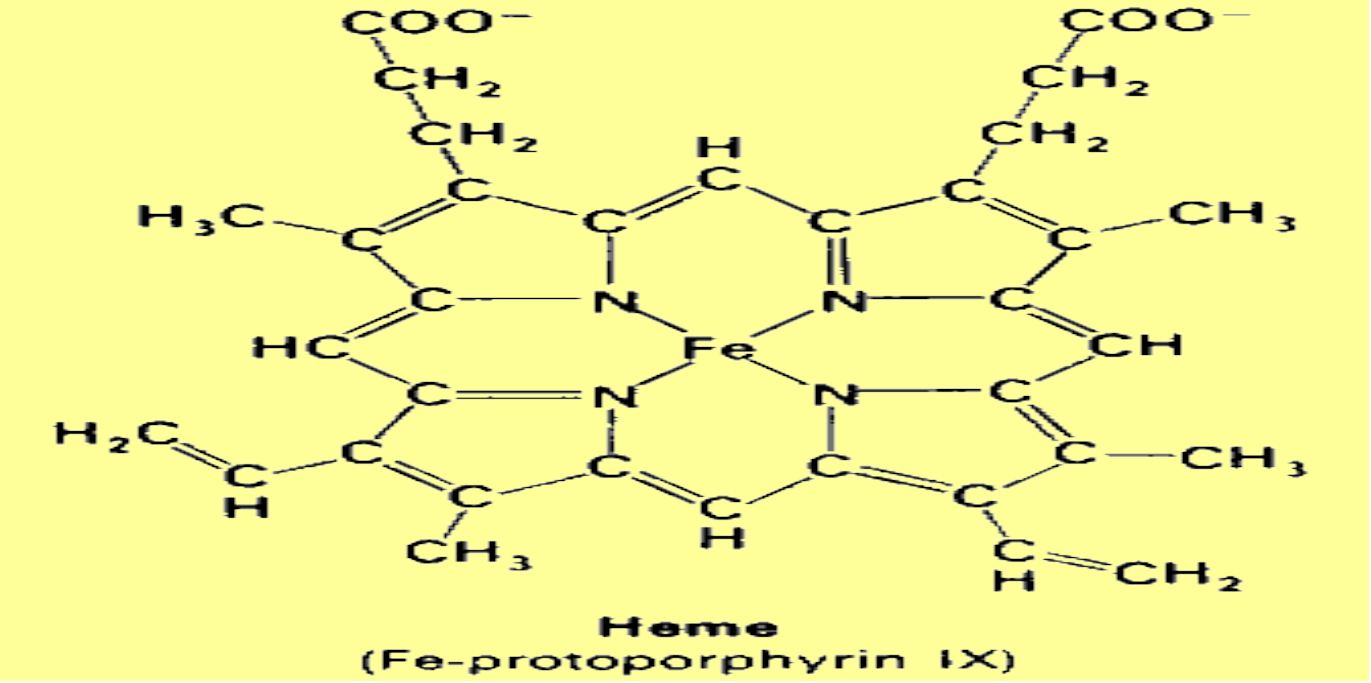
- Red blood pigment – found in erythrocytes.
- Can bind upto 4, $O_2$ molecule
- Consists of four polypeptide chains - 2**α** chains (141 amino acids/chain) & 2**β** chains (146 amino acids/chain).
- Unpaired globin precipitates, forming inclusions that damage the cell.
- Bind $O_2$ efficiently & retain at high $O_2$ tension (alveolus).
- Release at low O2 tension(tissue).
- $HbA_1$ has a total of 574 amino acid residues.

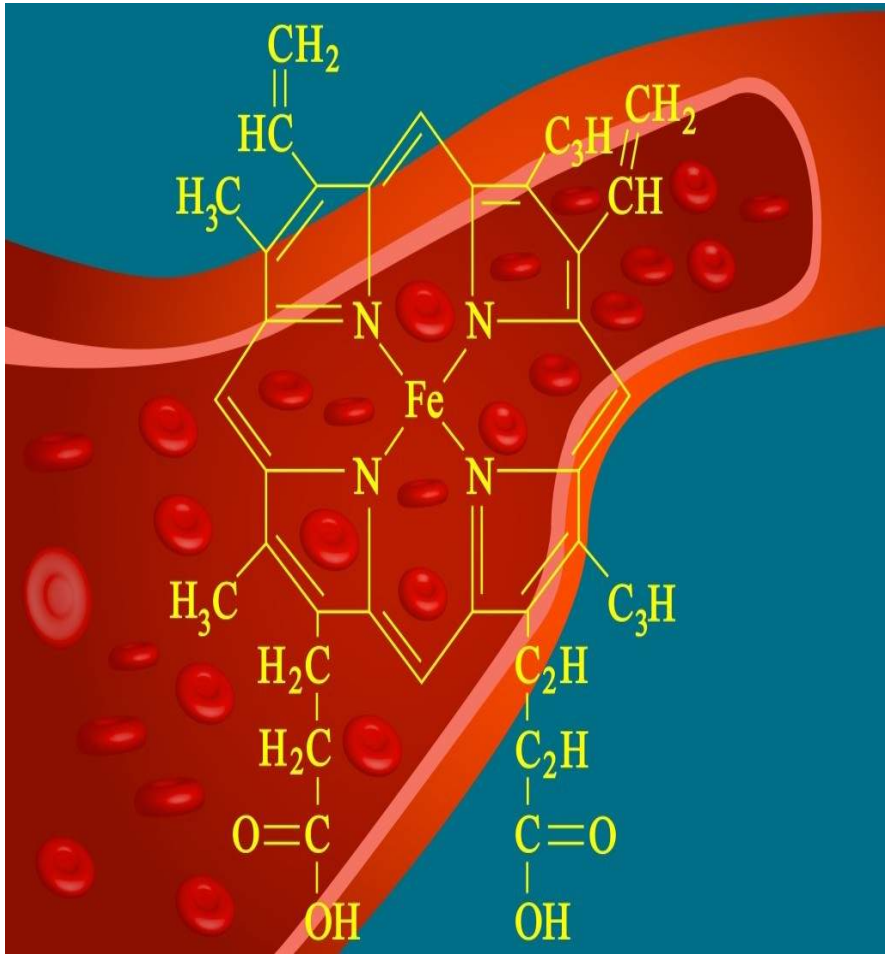
# Structure of haemoglobin




Each erythrocyte (RBC) contains ~270 million haemoglobin molecules

- The four subunits of HbA1 are held together by non-covalent interactions (hydrophobic, ionic and hydrogen bonds).
- The characteristics red coloration is due to heme.
- Heme – porphyrin molecule.
- Porphyrin – ***Protoporphyrin IX***



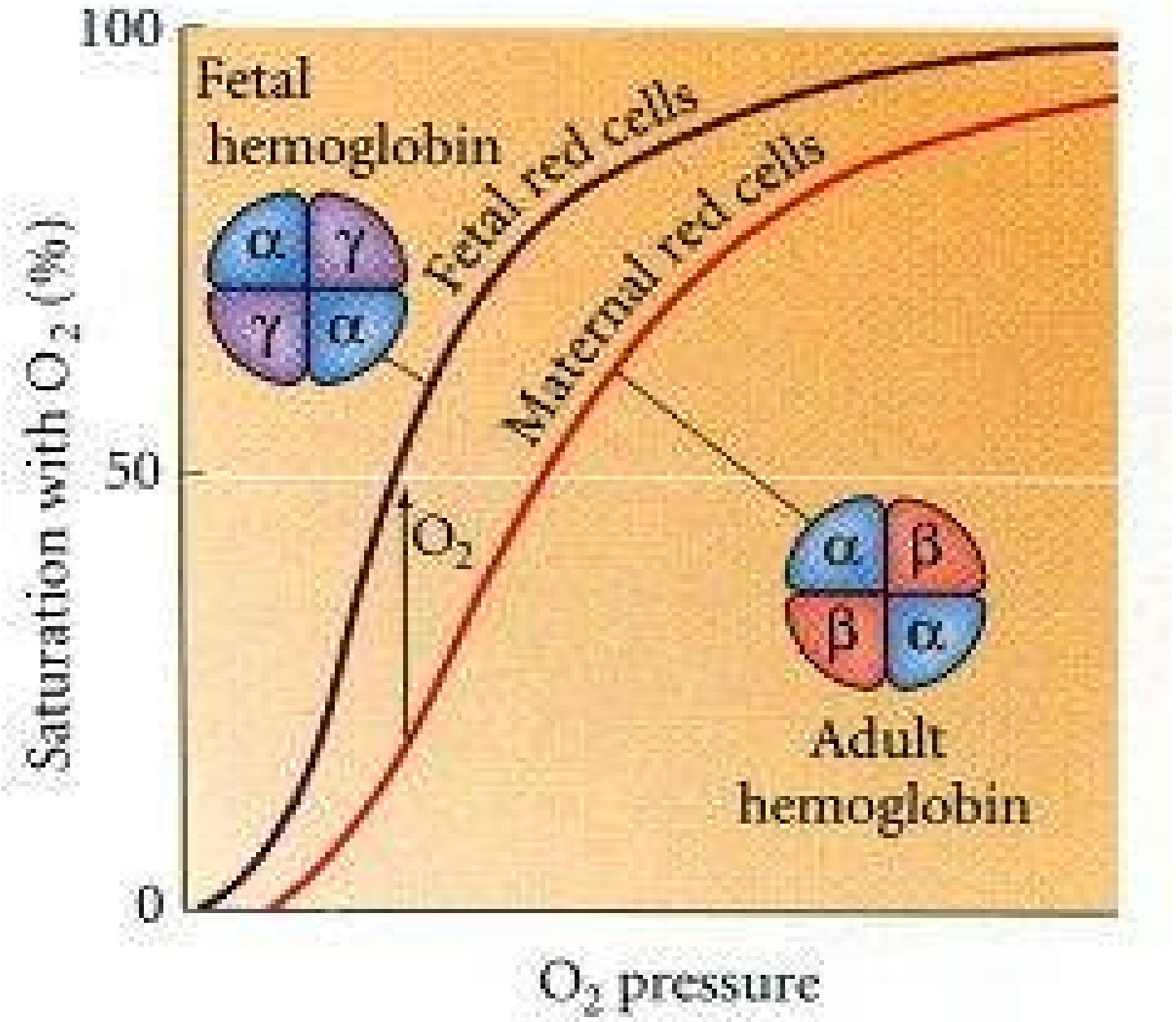



# Compositions of Normal Human Hb

| Type | Composition | Percentage of Hb |
|---|---|---|
| **$HbA_1$** | **$\alpha_2\beta_2$** | **90%** |
| **$HbA_2$** | **$\alpha_2\delta_2$** | **<5%** |
| **HbF** | **$\alpha_2\gamma_2$** | **<2%** |
| **$HbA_{1c}$** | **$\alpha_2\beta_2$ - Glucose** | **<5%** |

# FETAL HB

- It has $\alpha_2\gamma_2$ chains
- It has high $O_2$ affinity
- As a consequence, fetal hemoglobin can bind oxygen more efficiently than can adult hemoglobin. This small difference in oxygen affinity mediates the transfer of oxygen from the mother to the fetus
- Within the fetus, the myoglobin of the fetal muscles has an even higher affinity for oxygen, so oxygen molecules pass from fetal hemoglobin for storage and use in the fetal muscles
- In the placenta there is net flow of $O_2$ from mother to fetus


100
50
0
Saturation with $O_2$ (%)
Fetal hemoglobin
α γ
γ α
Fetal red cells
Maternal red cells
$O_2$
α β
β α
Adult hemoglobin
$O_2$ pressure

# HAEMOGLOBINOPATHIES

- The hemoglobinopathies are inherited disorders of hemoglobin synthesis (thalassemia) or structure (sickle cell disorders) that are responsible for significant morbidity.
- These disorders result in errors in oxygen-carrying capacity of hemoglobin .
- Genetic manifestation of Hb, instantly not only lethal, but resultant in long years of ill health and disability, etc.

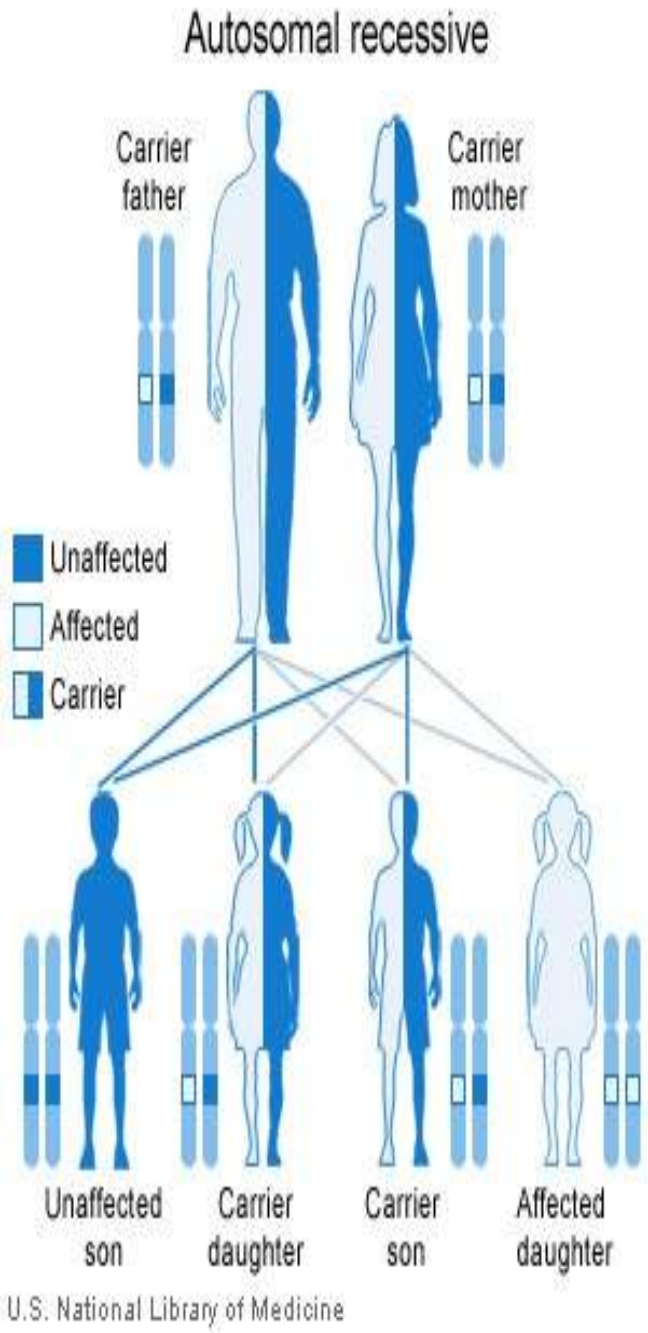
- Abnormally low levels of healthy hemoglobin or red blood cells is called anemia.
- Not enough hemoglobin, OR not formed correctly, leads to inefficient carrying of oxygen through the body.
- Depending on how much abnormal hemoglobin is being produced, symptoms can range from mild to life - threatening.
- ***Sickle cell anemia*** and ***thalassemia*** are examples of hemoglobinopathies.

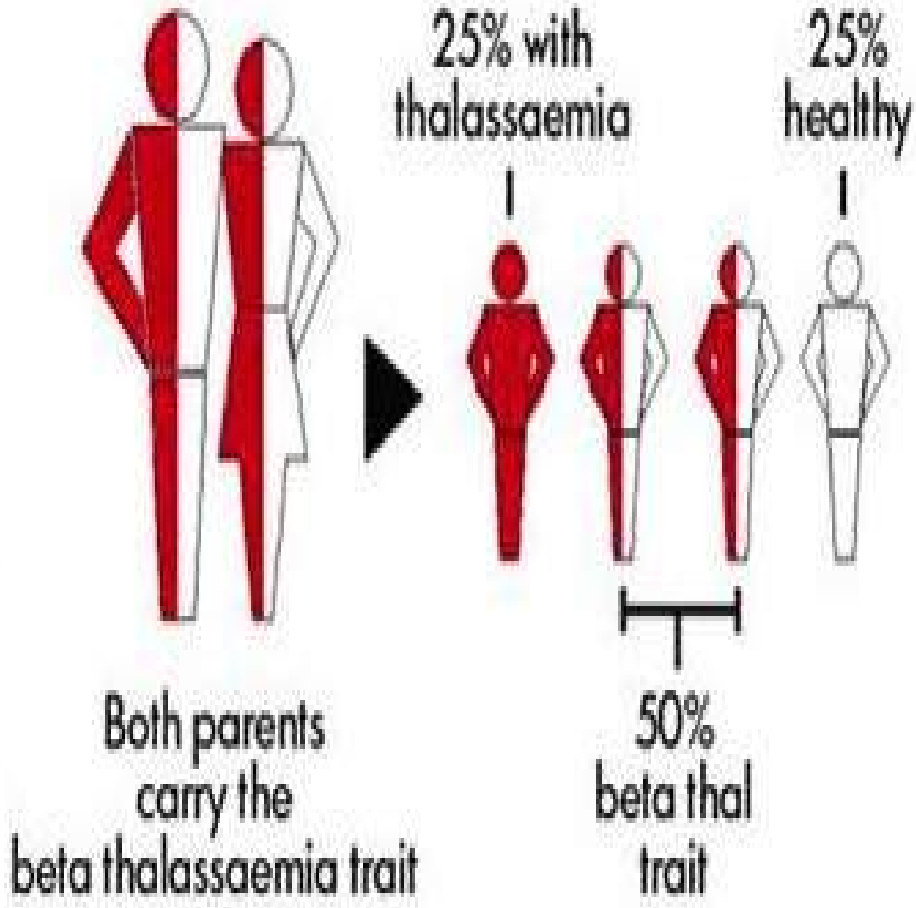
# Types/classification of Hemoglobinopathies

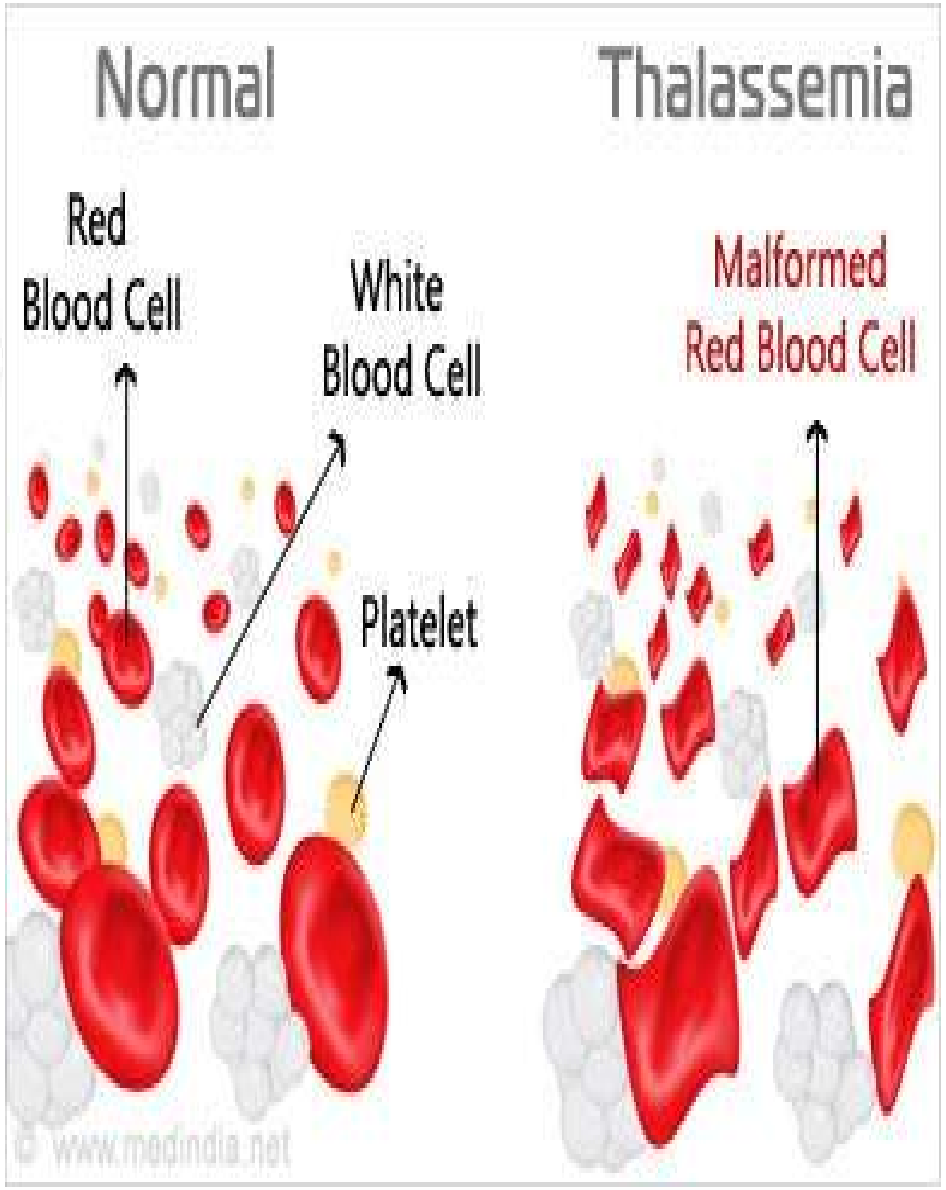
1) ***Structural hemoglobinopathies*** **-** hemoglobins with altered amino acid sequences e.g., HbS
2) ***Thalassemia*** **-** defective biosynthesis of globin chains.
3) ***Thalassemic hemoglobin variants*** **-** structurally abnormal Hb associated with co-inherited thalassemia phenotype
4) ***Hereditary persistence of fetal hemoglobin***
5) ***Acquired hemoglobinopathies***
    A. Methemoglobin
    B. Sulfhemoglobin
    C. Carboxyhemoglobin

# Hemoglobinopathies - Inheritance

- Hemoglobinopathies are inherited as ***autosomal recessive conditions.***
- Most people have two complete sets of genes, one inherited from each of our parents
- Individuals with one changed gene copy for a hemoglobin disorder are called carriers, or are said to have the trait.


Autosomal recessive
Carrier father
Carrier mother
Unaffected
Affected
Carrier
Unaffected son
Carrier daughter
Carrier son
Affected daughter
U.S. National Library of Medicine

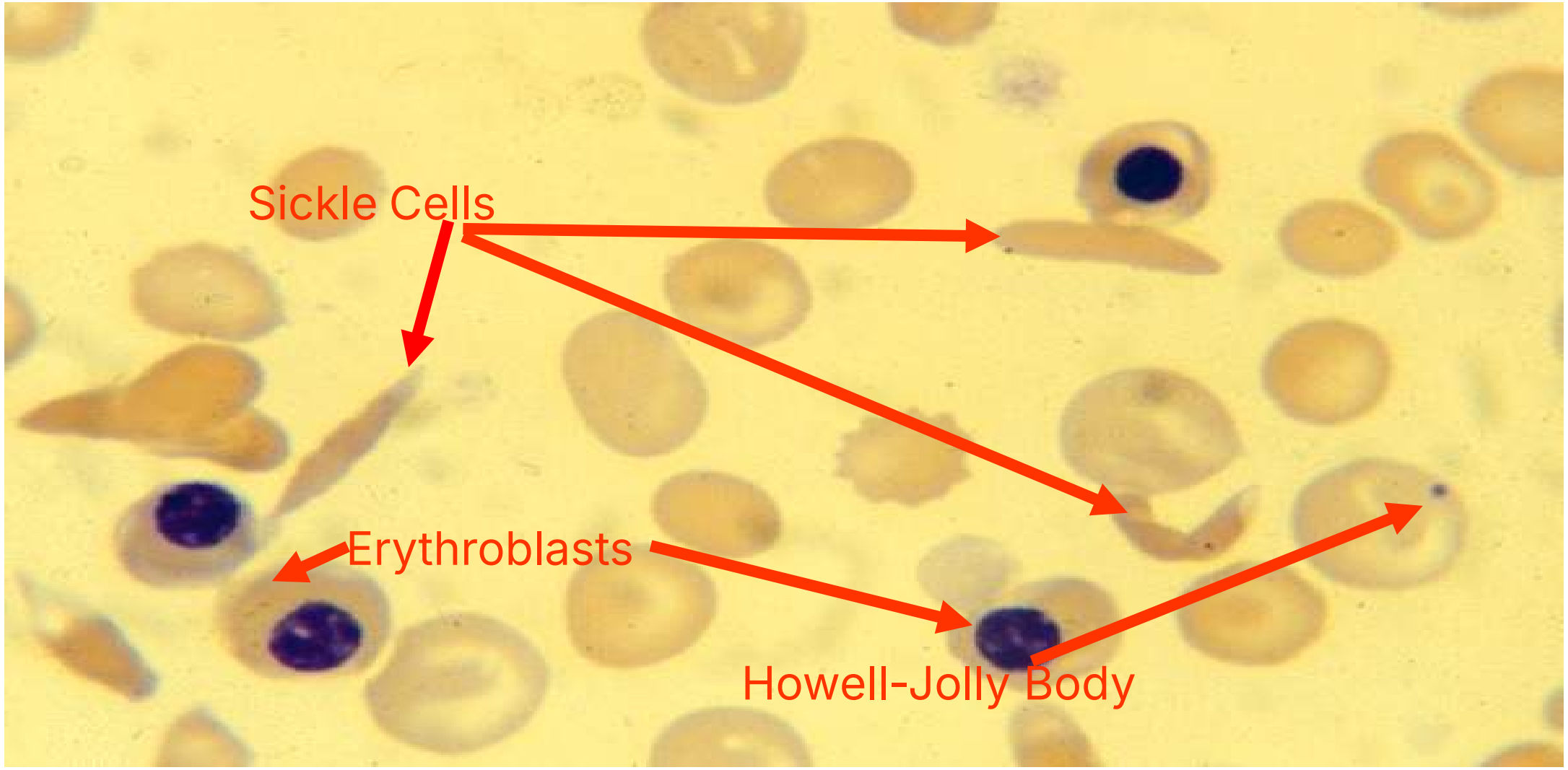

25% with thalassaemia
25% healthy
Both parents carry the beta thalassaemia trait
50% beta thal trait


Normal
Thalassemia
Red Blood Cell
White Blood Cell
Platelet
Malformed Red Blood Cell
© www.medindia.net

# Sickle Cell Anemia (HbS)

- A serious condition in which most of the red blood cells are sickled (crescent-shaped), instead of their normal round shape, at low oxygen concentration.
- This sickle shape is the result of a mutation in the beta globin genes.
- The abnormal shape is less efficient at carrying oxygen.
- In addition, sickled cells have a tendency to get stuck in smaller blood vessels, causing blockages.
- People with sickle cell disease tend to get tired easily, have episodes of pain, called “sickle crises”, and are prone to fevers and infections.

# Sickle Cell Anemia



# Molecular Biochemistry of Sickle Cell Anemia

- In SCA – has two normal alpha chains.
- Two mutant (abnormal) beta chains.
- This is mainly due to the difference in single amino acid residue, containing a ***Valine*** instead of a ***Glutamine*** at $6^{th}$ position.
- **Homozygous & Heterozygous - HbS**

# Abnormalities linked with HbS

1. **Life Long Hemolytic anemia: SC (**erythrocytes) are fragile, persistent breakdown leads to (chronic) life-long anemia.
2. **Tissue Damage and Pain:** The sickled cells block the capillaries resulting in poor blood supply to tissues. This leads to extensive damage and inflammation of certain tissues causing pain.
3. **Increased susceptibility to infection:** Hemolysis and tissue damage are accompanied by increased susceptibility to infection and diseases.
4. **Premature Death:** Homozygous individuals of SCA to die before they reach adulthood (<20years).

# What actually cause Sickling of Erythrocytes?

- Glutamate (Glu) is a polar amino acids. Glu is replaced by non-polar amino acid Valine in SCA.
- Valine – markedly decreases the solubility of HbS in deoxygenated form.



Individual homozygous for sickle-cell allele

↓

Sickle-cell (abnormal) hemoglobin

↓

Abnormal hemoglobin crystallizes when oxygen content of blood is low, causing red blood cells to become sickle shaped

Sickled cells



- Breakdown of red blood cells
  - Physical weakness
  - Anemia
  - Heart failure
- Clumping of cells and clogging of small blood vessels
  - Pain and fever
  - Brain damage
  - Damage to other organs
- Accumulation of sickled cells in spleen
  - Spleen damage

Anemia →
- Impaired mental function

Brain damage →
- Paralysis

Damage to other organs →
- Pneumonia and other infections
- Rheumatism
- Kidney failure

# Thalassemia

- Genetically linked moderate to severe anemia, mainly due to defect (impairment / imbalance) is reduced production of selected globin chains.

There are two basic groups of thalassemia.

α thalassemia: There are four types categorized according to the severity of their effects on persons with thalassemia.

ß thalassemia: There are 3 types categorized according to severity

- Thalassemia minor
- Thalassemia intermedia
- Thalassemia major

# α-Thalassemia

- An absence or deficiency of α-chain synthesis due to deletion of α-genes.
- Predominant cause of alpha thalassemia is large number of gene deletions in the α-globin genes on chromosome no. 16
- There are four clinical syndromes present in alpha thalassemia:

1. Silent Carrier State
2. Alpha Thalassemia Trait  (Alpha Thalassemia Minor)
3. Hemoglobin H Disease
4. Bart's Hydrops Fetalis Syndrome

# Variants of α-Thalassemia

## *1. Silent carrier*

Deletion of single α-gene
No. physical manifestation. Asymptomatic
Absence of RBC abnormality
Can only be detected by DNA studies.

## *2. Thalassemia trait*

Also called Alpha Thalassemia Minor.
Deletion of 2 α-genes
Minor anemia
Asymptomatic,
Minimal RBC abnormalities

# 3. Hemoglobin H Disease

- Second most severe form alpha thalassemia.
- Deletion of  3 α-genes
- 75% reduction of α-chain
- Only 25% α-chain synthesis small amount of HbF, HbA, & HbA2
- Fetus can survive
- Severe anemia
- Severe RBC abnormalities
- RBCs are microcytic, hypochromic

# 4. Bart’s Hydrops Fetalis Syndrome

Lack of all four genes.
Most severe form.
Incompatible with life.
Baby born with hydrops fetalis, which is edema and ascites caused by accumulation serous fluid in fetal tissues as result of severe anemia.
Also hepatosplenomegaly and cardiomegaly.

# β Thalassemia

- An absence or deficiency of β-chain synthesis of adult HbA.
- The molecular defects in β thalassemia result in the absence or varying reduction (according to the type of mutation) in β chain production.
- In individuals with beta thalassemia, there is either a complete absence of β globin production ( β-thalassemia major) or a partial reduction in β globin production ( β-thalassemia minor).

***Silent carrier state*** **-** the mildest form of beta thalassemia.

***Beta thalassemia minor*** **-** heterozygous disorder resulting in mild hypochromic, microcytic hemolytic anemia.

***Beta thalassemia intermedia*** **-** Severity lies between the minor and major.

***Beta thalassemia major*** **-** homozygous disorder resulting in severe transfusion-dependent hemolytic anemia.

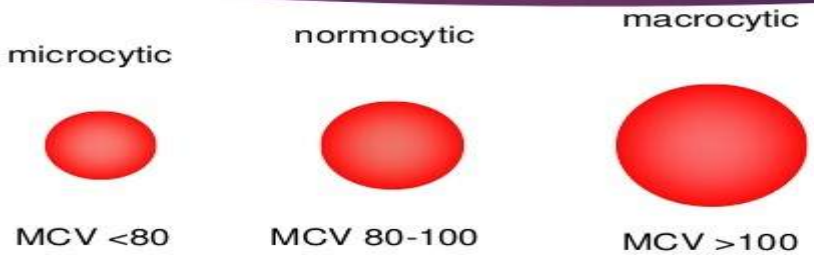
# Beta Thalassemia Minor

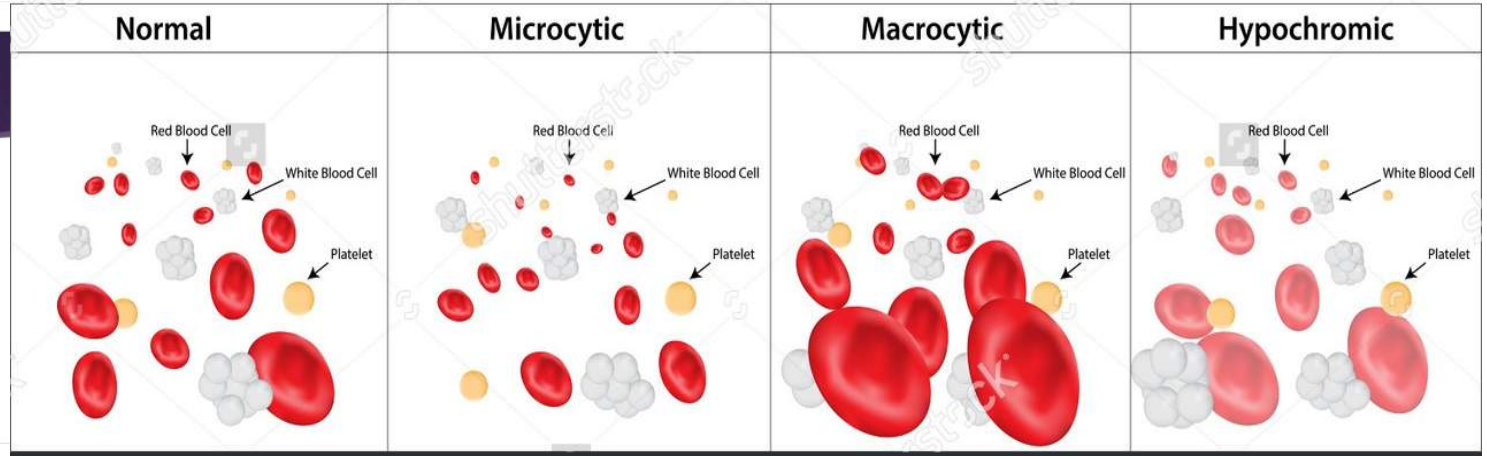
- Caused by heterogeneous mutations that affect beta globin synthesis.
- Usually presents as mild, asymptomatic hemolytic anemia unless patient in under stress such as infection or folic acid deficiency.
- Have one normal beta gene and one mutated beta gene.
- Anemia usually hypochromic and microcytic
- Normally require no treatment

# Beta Thalassemia Major

- Characterized by severe microcytic, hypochromic anemia.
- Detected early in childhood:
  - Infants fail to thrive.
  - Have pallor, variable degree of jaundice, abdominal enlargement, and hepatosplenomegaly.
- Hemoglobin level between 4 and 8 gm/dL.
- Severe anemia causes marked bone changes due to expansion of marrow space for increased erythropoiesis.

## Morphologic classification :




## RED BLOOD CELL MORPHOLOGY

| Size variation | Hemoglobin distribution | Shape variation | | Inclusions | Red cell distribution |
|---|---|---|---|---|---|
| Normal | Hypochromia 1+ | Target cell | Acanthocyte | Pappenheimer bodies (siderotic granules) | Agglutination |
| Microcyte | 2+ | Spherocyte | Helmet cell (fragmented cell) | Cabot's ring | |
| Macrocyte | 3+ | Ovalocyte | Schistocyte (fragmented cell) | Basophilic stippling (coarse) | Rouleaux |
| Oval macrocyte | 4+ | Stomatocyte | Tear drop | Howell-Jolly | |
| Hypochromic macrocyte | Polychromasia (Reticulocyte) | Sickle cell | Burr cell | **Crystal formation** | |



# THANK YOU